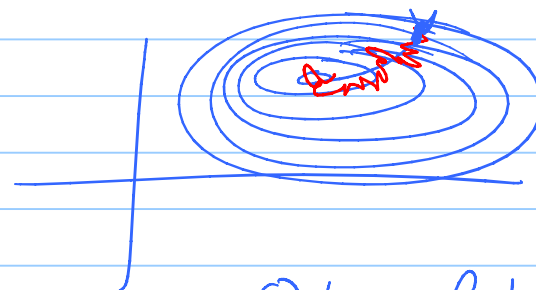


COL774

Marlon Lemg

Aug 13, 2024 -

Batch size = ∞ $\{x^i, y^i\}_{i=1}^m$ Shuffle $\{x^i, y^i\}_{i=1}^m$; $t \leftarrow 0$; $\mathcal{O}(t) \leftarrow \text{sort}()$; $k = 0 \neq 0$  f_i -Convex
guaranteed

① Lipschitz condition

$$\text{GD} \rightarrow f(y) \leq f(x) + \nabla f(x)^T (y-x) + \frac{L}{2} \|y-x\|^2$$

$$\text{SGD} \rightarrow \text{② } \text{Var}[\sqrt{\sum_{i=1}^n \epsilon_i}] \leq \sigma^2$$

no 2

Batch

$B \equiv$

$$\{x^{(b)}, y^{(b)}\}_{b=k \times \gamma + 1}^{(k+1) \times \gamma}$$

$$J^B(\theta)$$

$$\theta^{(k+1)} \leftarrow \theta^{(k)} - \eta \cdot \nabla_{\theta} J^B(\theta) |_{\theta^{(k)}}$$

$$\Rightarrow a = m/\gamma$$

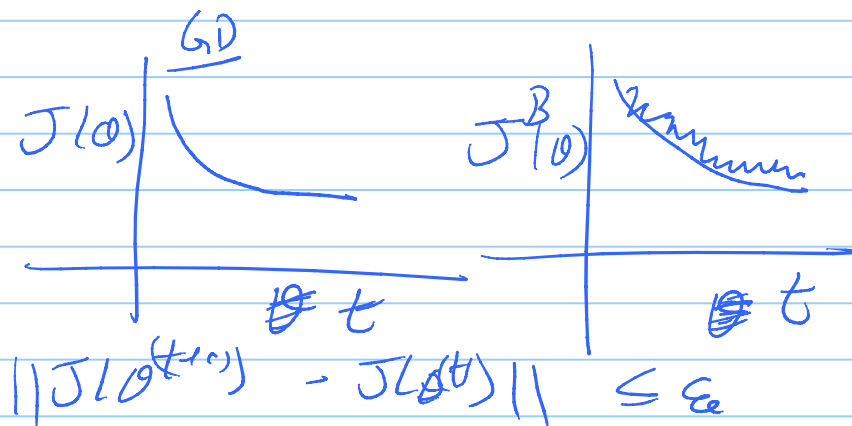


(epoch)

3 (whole, converged)

$$t \rightarrow t' \text{ to } t'+2 \quad J_{\text{avg}}^B(\theta^{(t)}) = \frac{1}{K} \sum_{k=1}^K J^B(\theta^{(t-k+1)})$$

$$\|J_{\text{avg}}^B(\theta^{(t')}) - J_{\text{avg}}^B(\theta^{(t)})\| \leq \epsilon$$



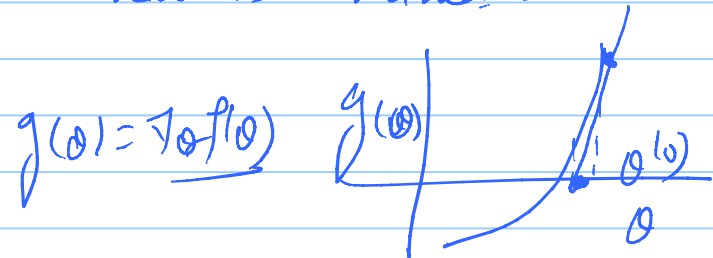
for 2 steps in continuous

① Newton's method (second order method) for optimization

② Probabilistic interpretation of least squares

③ Analytical solution of least squares

Newton's method -



$$\min_{\theta} f(\theta)$$

$$0: -J_0 f(\theta) = 0$$

↳ i.e.: iterative update

Newton's update -

$$\theta^{(t+1)} \leftarrow \theta^{(t)} -$$

$$J_0 f(\theta) = f(\theta^{(t+1)}) - f(\theta^{(t)})$$

$$\Rightarrow \theta^{(t+1)} = \theta^{(t)} - \frac{f(\theta^{(t)})}{J_0 f(\theta^{(t)})}$$